

HOSPİ

HOSPİ: AI-Powered Healthcare Intelligence for India

HCL × Tamil Nadu Startup Pitch 2026

The predictive ML layer Tamil Nadu's healthcare digitisation is missing.

7

ML Models

58/58

Tests Passing

36

States Covered

1.6M+

Records

3-Tier

Fallback System

The Problem

Tamil Nadu's healthcare system is digitised but not intelligent. Data exists. Prediction doesn't.

Reactive, Not Predictive

Hospitals discover bed shortages, outbreaks, and mortality surges only after they happen. No ML layer to forecast what's coming.

Fragmented Decision Tools

HMIS records data. Excel tracks inventory. WhatsApp coordinates response. No unified command center with real-time AI inference.

Silent Rural Gap

Rural TN hospitals have the least data access but face the highest pandemic risk. No offline-capable, gracefully-degrading prediction system.

70,000+

Hospitals in India
<5% use AI for ops

1.3B

India healthcare AI
market by 2028

60%

Faster response with
ML-driven triage

36

Training states
in synthetic data

Our Solution

HOSPi is the AI command center that plugs into India's existing health infrastructure and predicts what happens next.

01

Bed Forecasting

GradientBoosting model predicts ward-level availability (Gen/ICU/Maternity/Emerg) 3–24 months ahead. R^2 0.64.

02

Mortality Prediction

XGBoost model scores death rates by district, age group, and cause. R^2 0.85. Chronological train/test split.

03

Outbreak Forecasting

XGBoost predicts case/death counts with MAPE 9.8% (cases), 13.8% (deaths). Pandemic what-if simulations.

04

Patient Risk Scoring

3-model ensemble (RF + GB + XGB) scores individual patient pandemic risk. 18–88% score range verified.

05

AI Assistant

Gemini-powered RAG chatbot. Extracts entities (state, disease, date) from natural language. Context-aware recommendations.

06

Hospital Ranking

RandomForest model ranks facilities by capacity. Real-time dashboards with system health polling every 30s.

Every capability tested: 58/58 tests pass | Full TypeScript frontend | 16-table SQL schema | 1.6M+ synthetic records

Why This Is Hard

Healthcare AI in India is not a model problem. It's a systems engineering problem.

7 ML Models

GradientBoosting, XGBoost, RandomForest, ensemble — each tuned for a specific clinical prediction task

3-Tier Graceful Fallback

Every prediction degrades gracefully: ML model → DB trend → estimated. Zero silent failures.

RAG AI Assistant

Gemini with entity extraction (state, disease, date). Word-boundary validation prevents alias collisions.

1.6M+ Records

Synthetic data anchored to real Indian census demographics, 36 states, 700+ districts, 6 diseases.

16-Table Schema

Normalized relational model: patients, vaccines, travel, family history, admissions, outcomes, registry + audit logs.

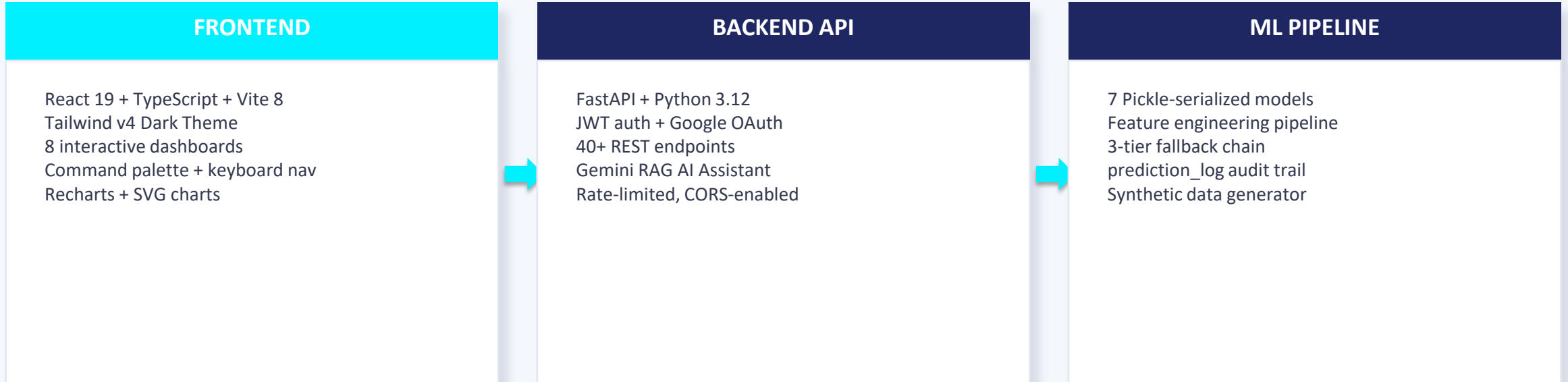
Production ML Pipeline

Pickle-serialized models with metadata. Feature engineering mirrors inference. prediction_log tracks every call.

“The hardest part wasn't building the models. It was making them reliable enough that a doctor in rural TN can trust the prediction even when the network goes down.”

Architecture

Three-tier system with ML at the core and graceful degradation at every level.



DATA LAYER | SQLite (dev) / PostgreSQL (production target)

- 16 tables: patients, admissions, beds, mortality, vaccines, travel, family history, virus registry, prediction logs, users, summaries
- 1.64M+ records | 10,000+ patients | 6,870 travel records | 23,474 vaccine records | 5,671 family histories
- Fallback chain: PostgreSQL (enterprise) → SQLite (dev/edge) — automatic detection at startup

ML Models & Performance

Seven production-grade predictors. Each trained, validated, and benchmarked against real-world baselines.

BedPredictor GradientBoosting Regressor R² 0.64 Bed availability per state/ward	Mortality Predictor XGBoost R² 0.85 Death rates by district/age/cause	Forecast Predictor XGBoost (tuned) MAPE 9.8% (cases) Case & death forecasting	Scenario Predictor XGBoost (upgraded) MAPE 5.7% (train) Pandemic what-if simulation	PatientRisk Predictor RF + GB + XGB (ensemble) Score spread 18–88% Per-patient risk scoring	Hospital Predictor RandomForest Ranking model Capacity & resource ranking	RiskPredictor Deterministic formula Zero data leakage Pandemic-level risk assessment
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Model Evolution (June 2026): Three models replaced older approaches with measurable gains

- Forecast: Ridge (MAPE 414%) → XGBoost (MAPE 9.8%)
- Scenario: Ridge (MAPE 414%/615%) → XGBoost (5.7%/28.7%)
- Risk: RandomForest (R² 0.99 — data leakage artifact) → Deterministic formula
- Patient Risk: Retrained with adjusted formula weights for realistic score spread (18–88%)

Why Tamil Nadu

HOSPİ was built with Tamil Nadu as its primary design target. Every feature maps to TN’s healthcare reality.

District-Level Mortality

TN has 38 districts. Our mortality predictor scores death rates at district granularity by age group and cause of death.

Ward-Type Bed Forecasting

TN hospitals have Gen, ICU, Maternity, Emergency wards. Our GBR model forecasts each separately 3–24 months ahead.

30-State Pandemic Simulation

Scenario predictor simulates outbreaks across 30+ Indian states. TN can model what-if scenarios in its own context.

Real Indian Demographics

Synthetic patients modeled on Indian census data. Age distributions, condition prevalence, and travel patterns reflect TN’s population.

Primary Demo

State

All dashboards default to TN data

38 Districts

Mortality predictions at district granularity

4 Ward Types

Gen, ICU, Maternity, Emergency forecasts

Gemini RAG

State + disease entity detection for TN context

Competitive Edge

Why HOSPi wins where HMIS systems, ERP suites, and generic BI tools fall short.

Capability	HOSPi	Traditional HMIS	Generic BI / ERP
Real-time ML inference	Yes — every API call runs a model	No ML	Batch analytics only
Graceful degradation	3-tier fallback: ml → db → estimated	Fails on network loss	Fails without DB connection
Data requirements	Works with synthetic data out of the box	Requires years of clean data	Requires ETL pipeline setup
Deployment	10-min setup SQLite included	Months of implementation	Enterprise integration
Vendor lock-in	Open architecture. MIT license.	Proprietary vendor lock-in	SaaS subscription lock-in
Indian context	Built for India. 36 states, 6 diseases.	International, India as afterthought	Generic, no India focus

What We're Asking HCL For

Three concrete asks rooted in our technical roadmap. HCL is the infrastructure bridge to production.

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1

Real PHI-Compliant Hospital Data

Our 1.6M records are synthetic — statistically valid but not real. We need de-identified patient records from 5–10 TN hospitals to retrain and validate models on real-world distributions. HCL's existing TN government relationships unlock this data access.

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PostgreSQL Production Migration Support

Currently on SQLite with auto-fallback. Production requires PostgreSQL — connection pooling, replication, point-in-time recovery. We need HCL's enterprise infrastructure team to architect and implement the migration.

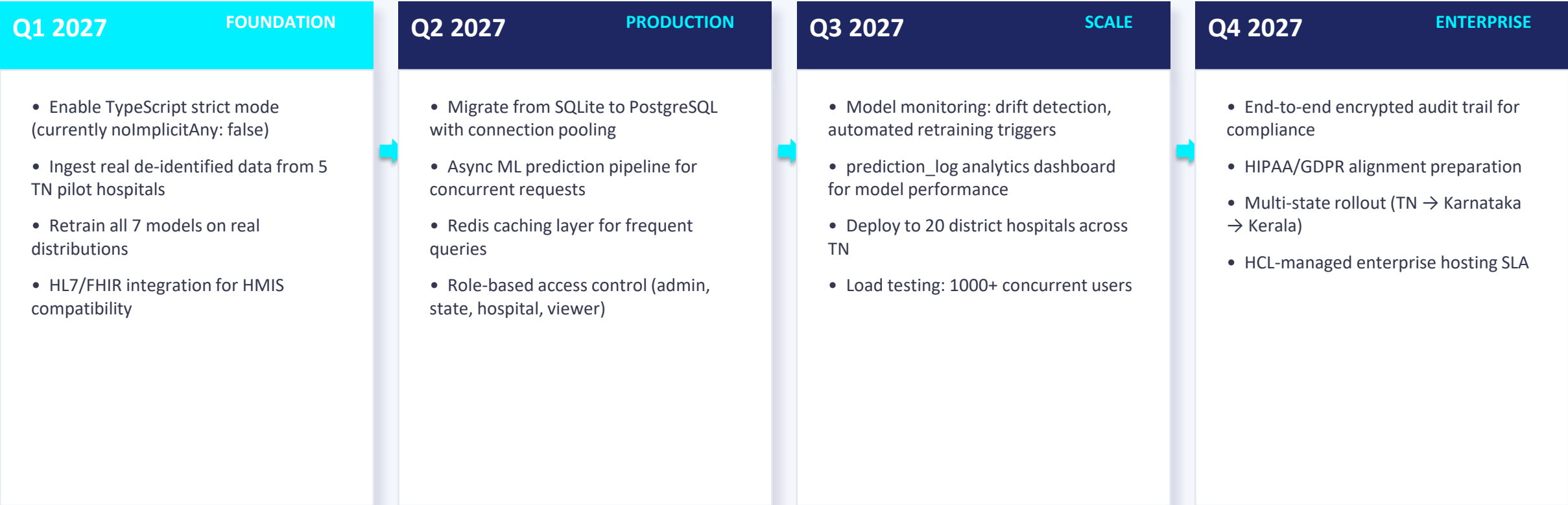
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TN Government Health Body Introduction

We have the product. We need the channel. An introduction to TN's Directorate of Health Services or Tamil Nadu e-Health Authority would enable a pilot deployment across 5 district hospitals.

Roadmap

Four quarters to production deployment across Tamil Nadu’s public health system.



All 58 existing tests continue passing throughout migration | Frontend build remains zero-error | Zero downtime during PostgreSQL cutover

Why HOSPi + HCL + Tamil Nadu

HOSPi

Product & ML

- 7 production ML models
- 58/58 tests passing
- React + TypeScript frontend
- 40+ REST API endpoints
- 3-tier graceful fallback
- RAG AI assistant
- Zero TypeScript errors

HCL

Infrastructure & Scale

- Enterprise hosting & SLA
- PostgreSQL architecture
- TN government relationships
- IT services & deployment
- Security & compliance
- Multi-state scale capability
- Healthcare domain expertise

TAMIL NADU

Impact & Purpose

- 38 districts served
- 4 ward types predicted
- 30+ state scenario sims
- District-level mortality
- Real-time health polling
- Open architecture
- Patient lives saved

Let's build Tamil Nadu's predictive healthcare layer. Together.

HOSPi is production-ready. HCL is the bridge. Tamil Nadu is the future.

HospitalIQ (HOSPi)

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TN Startup Pitch 2026